2ai	First Law of thermodynamics states that the increase in internal energy of a system is equal to the sum of heat supplied to the system and the work done on the system.	B1
aii	Small volume change/decrease and hence work done on ice is negligible/positive, thermal energy is absorbed to break lattice structure / increase molecular potential energy. Hence, by first law of thermodynamics, internal energy increases.	B1 B1
bi	Constant volume, therefore W = 0 J $\Delta U = Q + W$ $\Delta U = Q$	C1
	$\Delta U = -55$ J	A1
bii	Process A to B takes place at constant temperature, therefore $U_A = U_B \Rightarrow \Delta U_{AB} = 0$	C1
	For cyclic process, $\Delta U_{ABCA} = 0$ $\Delta U_{AB} + \Delta U_{BC} + \Delta U_{CA} = 0$ $0 + (-55) + \Delta U_{CA} = 0$ $\Delta U_{CA} = 55$	C1
	Process C to A is an adiabatic process, therefore $Q_{CA} = 0\,J$ $\Delta U_{CA} = Q_{CA} + W_{CA}$ $55 = 0 + W_{CA}$ Since WD on gas from C to A = 55 Therefor WD by gas from C to A = -55 J	A1
	Total:	**8**